

Reading Group Applied Econometrics and ML

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A gentle Introduction to CATE

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Pre-requisite to digression DR

- **Causal Inference:** Understanding how a treatment (e.g., policy, intervention) affects an outcome when randomized control trials (RCTs) are not feasible. Essential in observational studies to differentiate **causation** from **correlation**.
 - **Challenge:** Confounding variables (variables that influence both the treatment and the outcome) create biases in observational data, making the estimation of true causal effects difficult.
 - **Methods:** Matching and regression adjustments aim to replicate RCT conditions.
- **Average Treatment Effect (ATE):**
 - Uses the **Rubin-Neyman causal model** of potential outcomes.
 - Measures the difference in average outcomes between the treated and control groups:

$$\text{ATE} = \mathbb{E}[Y_1 - Y_0] = E_x[\mu_1(X) - \mu_0(X)]$$

- **Key Issue:** We cannot observe both outcomes (treated and untreated) for the same unit, but assumptions can enable meaningful inferences.

The issue is usually solved by assuming:

① **SUTVA**:

- The treatment received by one unit does not impact the outcomes of other units.
- A unit's outcome depends solely on the treatment it receives, not on the method of delivery.

② **CIA** (ignorability): after controlling for all the relevant factors¹, the only difference between the people who got the treatment and those who didn't should be the treatment itself:

$$Y_i(1), Y_i(0) \perp\!\!\!\perp D|X$$

③ **Common Support**: units in both the treatment and control groups have a comparable range of covariates ²:

$$0 < Pr(D = 1|X) < 1$$

¹so the more variables we control for the better

²which is more likely the less X s we have (contrast with CIA)

ATE via Outcome modeling:

- **Objective:** Estimate the effect of a treatment by predicting outcomes under different scenarios using observational data.
- **Average Treatment Effect (ATE):** Measures the average difference in outcomes between treated and untreated groups.

$$\text{ATE} = E[Y_1 - Y_0] = E[E[Y_1 | X, D = 1] - E[Y_0 | X, D = 0]]$$

- **Using a Function μ :**

- We approximate the expected outcome with a function $\hat{\mu}_t$ as:

$$\mu_t(X) = E[Y_t | D = d, X] + \epsilon_t$$

- This function $\hat{\mu}$ helps us predict outcomes based on characteristics X and treatment status D .
- **Computing ATE:**

$$\text{ATE} = \frac{1}{n} \sum_i (\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i))$$

- **Challenges:** Ensuring overlap between treated and untreated groups, and correctly specifying the model to avoid biased estimates.

ATE via IPW (1):

- **Definition:** Propensity scores represent the probability that a unit receives a specific treatment based on its observed characteristics:

$$p(X_i) = \Pr[D = 1 \mid X_i]$$

- **Purpose:** To create comparable groups by adjusting for confounding variables and to facilitate simpler comparisons.
- **Methods using Propensity Scores:**
 - **Matching:** Pairing treated and control units with similar propensity scores.
 - **Stratification:** Dividing data into subgroups based on similar propensity scores.
- **Fallacies:**
 - **Incomplete Matching:** Difficulty in finding suitable matches for all treated units, leading to biased estimates.
 - **Imperfect Stratification:** When propensity scores are not well estimated, remaining confounding effects can still bias the results.

ATE via IPW (2):

- **Definition:** IPW is a method that re-weights observations based on the inverse probability of receiving the observed treatment.

The intuition comes from observing that $\widehat{ATE} := E_n[Y|D = 1] - E_n[Y|D = 0]$ can be re-written as a function of propensity score noticing that:

$$E \left[\frac{Y_i(d)1\{D_i = d\}}{p(X_i)} \mid X \right] = E[Y_i(d) \mid X_i]$$

- **ATE Calculation Formula** Specifically...(see Callaway, Sant'Anna; 2021)

$$\begin{aligned} \widehat{ATE} &= \frac{1}{n_1} \sum_i \frac{Y_i(d)1\{D_i = 1\}}{\hat{p}(X_i, W_i)} - \frac{1}{n_0} \sum_i \frac{Y_i(d)1\{D_i = 0\}}{1 - \hat{p}(X_i)} \\ &\rightarrow E_n \left[\frac{D_i}{p(X_i)} Y_i \right] - E_n \left[\frac{(1 - D_i)}{1 - p(X_i)} Y_i \right] \end{aligned}$$

- **Intuition:** IPW gives more weight to individuals with uncommon characteristics to balance treated and untreated groups. This helps mitigate biases arising from incomplete matching or misaligned propensity scores.

In both methods (and in DR as we will see) **the Y is unconstrained**, i.e. it can assume any functional form! For instance, one of the simpler is that of Partial Linear Model (PLR):

$$Y = \underbrace{\alpha}_{\text{elasticity}} \cdot D + \underbrace{f_0(X)}_{\text{effect of 1000s potential confounders}} + \varepsilon$$

PLR, however, implies homogeneous effects (there is a unique coefficient of interest to estimate, α , such that $E[\alpha|X] = \alpha$, i.e. is constant)! Instead, we want to estimate heterogeneous effects, meaning we want to estimate a different coefficient depending on X . So we would like that μ not to be constant across X but to depend on X , i.e. $\alpha_t(X)$ or functions of $\alpha_t(X)$!

Thus though Y is unconstrained not all functional forms are appropriate for estimating CATE!

Thus, in DR we will not make any functional assumption on Y , but we restrict on functions of Y that do not imply homogeneous effects!

We will **still focus on ATE estimators as above but stress more the fact that ATE is an average of CATE** (i.e. we will recover a $\phi(X) = f(\mu_X)$ where $\mu(X)$ is the equivalent of $\alpha(X)$ in an unconstrained model and estimate the ATE as $E[\phi(X)]$).

Wrapping up until now...

We wanna estimate the ATE as an average of CATE for generalization purposes:

- $\text{CATE} := E[Y_1|X = x] - E[Y_0|X = x] = \mu_1(X) - \mu_0(X)$
- CATE identification implies ATE identification by averaging across all the values of X :

$$\text{ATE} := E_x[\mu_1(X) - \mu_0(X)]$$

To do so we:

- Make no functional form assumptions on Y that could imply homogeneous effects (we want to estimate heterogeneous effects);
- Notice that CATE could be computed also via IPW (Abrevaya, 2015) and outcome modeling. The focus on DR is just to exploit the advantages of both IPW and outcome modeling methods (see below)

ATE via DR

- **Definition:** Double Robust (DR) estimation combines outcome modeling and IPW strengths. Gives a consistent estimate whenever either the outcome model or the propensity score model is correct.
- **Estimator Formula:**

$$\hat{Y}_d(X_i) = \begin{cases} \mu_d(X_i) & \text{if } i \text{ did not have treatment } t \\ \mu_d(X_i) + \frac{Y_i(d) - \mu_d(X_i)}{p_d(X_i)} & \text{otherwise} \end{cases}$$

- **Theorem:** The ATE is given by:

$$E[Y_1 - Y_0] = E[\mu_1(X) - \mu_0(X)] = E[\hat{Y}_1 - \hat{Y}_0]$$

- **Outcome Model Correct:** If $\mu_d(X_i)$ is correct, the adjustment term $\frac{Y_i(d) - \mu_d(X_i)}{p_d(X_i)}$ converges to zero ($Y_i(d) - \mu_d(X_i) = 0$), leaving $\hat{Y}_d(X_i) = \mu_d(X_i)$.
Propensity Score Correct: If $p_d(X_i)$ is correct (and outcome-modeling is not, e.g. due to absence of overlap), the weighting factor $\frac{1}{p_d(X_i)}$ effectively corrects for non-random assignment into treatment groups in $\mu_d(X_i)$, i.e. selection bias.

A digression on Doubly Robust (DR) methods

More formally we can speak about identification...

- The DR shows that under the CIA the identification of treatment effects can be based both on conditional mean outcomes $\mu_1(X)$ and $\mu_0(X)$ (outcome modeling), and based on weighting using the propensity score $p(X)$ (IPW)
- Indeed, ATE and ATT are **identified based on the above stated (re-written), DR, population moment conditions:**

$$ATE = E[\hat{Y}_1 - \hat{Y}_0] = E[\phi(X)],$$

$$\phi(X) = \mu_1(X) - \mu_0(X) + \frac{(Y - \mu_1(X)) \cdot D}{p(X)} - \frac{(Y - \mu_0(X)) \cdot (1 - D)}{1 - p(X)}$$

$$ATT := E[\phi(X)|D = 1] := \text{Law of total expect.}^3 \frac{E[\phi(X)D]}{\Pr(D=1)} =$$

$$E \left[\frac{(Y - \mu_1(X)) \cdot D}{\Pr(D=1)} - \frac{(Y - \mu_0(X)) \cdot (1 - D) \cdot p(X)}{(1 - p(X)) \cdot \Pr(D=1)} \right]$$

- $\phi(X)$ is called **efficient influence function** and as we will see in the next slide is a re-writing of the known $\mu_1(X) - \mu_0(X)$

³By law of total expectations, $E[Z] = E[Z|D = 1]P(D = 1) + E[Z|D = 0]P(D = 0)$: in our case $Z := \phi D$ and given that $D = 1$, $E[Z|D = 0]P(D = 0)$ cancels. Then we just recover $E[Z|D = 1]$ from $E[Z] = E[Z|D = 1]P(D = 1)$

A digression on DR methods

- The above expressions may seem more complicated than the population **moment conditions describing outcome modeling and IPW separately**, but **they coincide with them**.
- This follows from the fact that by the law of iterated expectations (LIE), some terms cancel out:

$$E \left[\frac{(Y - \mu_1(X)) \cdot D}{p(X)} - \frac{(Y - \mu_0(X)) \cdot (1-D)}{1-p(X)} \right] = E \left[\frac{\varepsilon \cdot D}{p(X)} - \frac{\varepsilon \cdot (1-D)}{1-p(X)} \right] = 0, \text{ i.e.}$$

$\phi(X)$ coincides with CATE: $\mu_1(X) - \mu_0(X)$ ⁴ and

$$E[\phi(X)] = E[\mu_1(X) - \mu_0(X)] := \text{ATE}$$

Similarly for ATT we can show that $\frac{E[\phi(X)D]}{Pr(D=1)}$ identifies $\frac{E[(\mu_1(X) - \mu_0(X))D]}{Pr(D=1)}$;

- with the error term $\varepsilon = Y - \mu_D(X)$ and $E[\varepsilon|D, X] = 0$

⁴Remember that LIE states $E[Z] = E[E[Z|X]]$: since it's $E[\varepsilon|D, X]$ equals 0, i.e. only after conditioning on X , the LIE is applied on $Z = \frac{(Y - \mu_1(X)) \cdot D}{p(X)} - \frac{(Y - \mu_0(X)) \cdot (1-D)}{1-p(X)}$

A digression on DR methods

- **Despite the equivalence of DR, IPW, and outcome regression** for identifying causal effects in the population **when assuming correct models** for the conditional mean outcomes and the propensity score, **DR has an attractive property** when it comes to estimation, which may be prone to misspecification.
- In fact, as above specified DR estimator based on the sample version of the previous population conditions with plug-in estimates for $p(X)$, $\mu_0(X)$ and $\mu_1(X)$ **is consistent if either the conditional mean outcomes or the propensity scores are correctly specified** (Robins, Mark, and Newey, 1992; Robins, Rotnitzky, and Zhao, 1995).
- Put differently, **DR remains consistent if either the propensity score or the conditional mean outcomes models are incorrect.**
- This makes DR estimation **more robust to model misspecification** than the other two methods, as suggested by its name, as we now have 'two shots' for getting either $p(X)$ or $\mu_0(X)$ and $\mu_1(X)$ right.

LASSO to Select Covariates and DR

Farrell (2015) and Chernozhukov et al. (2018) proposed that modern model selection techniques, such as LASSO methods for inference, be combined with the DR estimator to estimate the ATEs. The intuition behind this method is twofold:

- On one hand, **LASSO techniques resolve the conflicts between the CIA and overlap assumptions** (see above).
- Although CIA expects many variables, we only need the covariates that matter for the outcome and the treatment.
- If LASSO selects a sparse model, the overlap assumption is more plausible to be satisfied.
- On the other hand, the proposed estimator should **guard against model selection mistakes**.
- The DR estimator happens to satisfy this property. **Double robustness** means that the estimates of treatment effects are consistent if either the outcome model or the treatment model is correctly specified.

LASSO is cool (solves CIA vs overlap trade-off, allows for tons of covariates via sparsity), but comes with a price: regularization bias.

Chernozhukov et al. (2020) show that the inference is still valid, specifically, the inference is robust to the mild model selection mistakes that could happen (e.g. misspecification of $f_0()$ in **Example 1** below)⁵

In their influential paper, the authors show that the estimator is still consistent even if model selection makes some mild mistakes **provided that the moment conditions on which it is based are Neyman orthogonal**.

This point is important since it allows us to perform valid estimation and inference in a general class of models where confounders are either nonparametric or linear and high-dimensional...

⁵Notice: in ML jargon for "model" we do not mean the structural form as corrected via DR instead, loosely speaking, in a LASSO linear model "the number of X s included". In MR, e.g. $Y = \alpha + \beta_1 X_1 + \varepsilon$ is a different "model" than $Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$

We reached a somehow crucial point in causal ML estimation. Let's try to get the point visually. PLR is used here for illustrative purposes:

Example 1

$$Y = \underbrace{\alpha}_{\text{elasticity}} \cdot D + \underbrace{f_0(X)}_{\text{effect of 1000s potential confounders}} + \varepsilon$$

Problem: Accurate estimation of nuisance function $f_0()$ often unachievable when $f_0(.)$ nonparametric or linear and high-dimensional;

Assuming Neyman orthogonality allows us to do the following:

- 1 We are ensured that even if we estimate $f_0()$ ⁶ poorly...
- 2 the **Neyman orthogonal** estimator of α is robust to first order errors in estimation of nuisance $f_0() + \sqrt{n}$ -consistent

⁶P.s. As you see it is not questioned the functional form of Y in toto but the part of Y modeled via ML, i.e. $f_0()$. this is what we meant above by "model selection mistakes".

The DRL and a recap:

- So before we said that DR estimator identifies CATE and ATE;
- We then saw how LASSO can enter the DR framework to create an estimator that: 1) solves CIA vs overlap issue; 2) is robust to model misspecification; 3) does not restrict the functional form of Y ;
- This estimator to be valid needs to satisfy Neyman orthogonality. **Example 1** shows how, for instance, the assumption of Neyman orthogonality allows for valid estimation in a general class of models like PLR;
- But how is this estimator actually constructed? In the next slides we will see the general recipe for constructing such an estimator. The procedure is called Doubly Robust Learning (DRL). We then apply DLR to the DR specifically (via a moment function coinciding with $\phi(X)$)...

DRL $\simeq$ DML

- **Doubly Robust Learning** is a subclass of Double Machine Learning and is used for estimating (heterogeneous) treatment effects when:
 - The treatment is **categorical**.
 - All potential confounders/controls are **observed**.
- Is applied whe:
 - **Too many confounders** for classical methods.
 - **Non-parametric relationships** between treatment, confounders, and outcome.
- ML techniques can address these challenges (e.g., Chernozhukov, 2016; Foster, 2019).
- Originating from early works like Robinson (1994), this method was initially applied to estimate average treatment effects.
- Recent modifications allow for the estimation of **heterogeneous treatment effects, CATE** (e.g., Foster, 2019).

Orthogonalization and DRL : general recipe

The DRL procedure uses the same procedure as the DML one, i.e. a sort of GMM:

- **First input:** a score function $\psi(w_i, \alpha, \eta)$ such that:
 - ① $E[\psi(w_i, \alpha, \eta)]$ identifies the true α , α_0 when $\eta = \eta_0$, i.e. $E[\psi(w_i, \alpha, \eta_0)] = 0$ iff $\alpha = \alpha_0$;
 - ② the Neyman Orthogonality condition is satisfied, i.e.:

$$\partial_{\eta} E[\psi(w_i, \alpha, \eta)]|_{\eta=\eta_0} = 0$$

- **Second input:** a sufficiently powerful ML tool for estimating the nuisance parameters η , i.e. a ML model such that the moment condition $\omega(w_i, \hat{\eta}) = 0$ when $\hat{\eta}$ is estimated via such a ML tool;
- **Final input:** sample splitting procedure (cross-fitting) at the stage of producing the estimator for the main parameter α_0 allowing to avoid biases from overfitting and p-hacking (see later on).

Let's see how these ingredients are put together to form the estimator

Orthogonalization and DRL : First + second input

- Define α as parameters of interest and η as nuisance parameters⁷.
- The estimate $\hat{\alpha}$ is obtained following first step estimate of $\hat{\eta}$ (e.g. PLM pag. 272 book Chern. for PLM example)

First stage: $\hat{\eta}$ solves $\sum_{i=1}^n \omega(w_i, \eta) = 0$

Second stage: $\hat{\alpha}$ solves $\sum_{i=1}^n \psi(w_i, \alpha, \hat{\eta}) = 0$, α true ATE (τ below).

where LASSO (ML in general) is used to estimate η and α .

- Results: first-stage estimation of η does not effect the second-stage asymptotic distribution of $\hat{\alpha}$ and small mistakes in η are unimportant if the second-stage function $\psi(\cdot)$ satisfies Neyman orthogonality

$$E [\partial \psi(w_i, \alpha, \eta) / \partial \eta] = 0$$

- Intuition: on average changing η does not change $\psi(\cdot)$

⁷coefficients or parameters in the model that are not of direct interest but need to be

Orthogonalization and DRL: Application to DR

- We want to estimate $\widehat{ATE} = \hat{\tau} = \hat{\mu}_1 - \hat{\mu}_0$ where

$$\hat{\mu}_j = \frac{1}{n} \sum_{i=1}^n \left\{ \frac{\mathbf{1}[d_i = j] (y_i - \hat{\mu}_j(\mathbf{x}_i))}{\hat{p}_j(\mathbf{x}_i)} + \hat{\mu}_j(\mathbf{x}_i) \right\}.$$

- The ATE solves the population moment condition (i.e. what is the ψ such that $E[\psi(\cdot)] = 0$?)

$$E \left[\frac{\mathbf{1}[d = 1] (y - \mu_1(\mathbf{x}))}{p(\mathbf{x})} + \mu_1(\mathbf{x}) - \frac{\mathbf{1}[d = 0] (y - \mu_0(\mathbf{x}))}{1 - p(\mathbf{x})} - \mu_0(\mathbf{x}) - \underbrace{\tau}_{\alpha} \right] = 0.$$

- The parameter of interest is τ
- Define the nuisance parameters on $\phi(X) := \mu_1(X) - \mu_0(X)$
 $\eta_1 = \mu_1(\mathbf{x}) = E[y_1 | \mathbf{x}]$, $\eta_2 = \mu_0(\mathbf{x}) = E[y_0 | \mathbf{x}]$ and $\eta_3 = p(\mathbf{x})$.
- Then $E[\psi(w, \tau, \eta)] = 0$ for

$$\psi(w, \tau, \eta) = \frac{\mathbf{1}[d = 1] (y - \eta_1)}{\eta_3} + \eta_1 - \frac{\mathbf{1}[d = 0] (y - \eta_2)}{1 - \eta_3} - \eta_2 - \tau.$$

Orthogonalization and DRL: Application to DR

- We have $\psi(\mathbf{w}, \tau, \eta) = \frac{1[d=1](y-\eta_1)}{\eta_3} + \eta_1 - \frac{1[d=0](y-\eta_2)}{1-\eta_3} - \eta_2 + \tau$
- This satisfies the orthogonalization condition

$$E[\partial\psi(\mathbf{w}, \tau, \eta)/\partial\eta_1] = E\left[-\frac{1[d=1]}{\eta_3} + 1\right] = 0$$

★ since $E[1[d=1]] = E[p(\mathbf{x})] = \eta_3 : (p(X)=Pr[D=1|X], \text{ so } E_x[p(X)] = Pr(D=1)).$

Hence, $E[p(X)]=Pr(D=1)=E[1[d=1]]$

$$E[\partial\psi(\mathbf{w}, \tau, \eta)/\partial\eta_2] = E\left[\frac{1[d=0]}{1-\eta_3} - 1\right] = 0$$

★ since $E[1[d=0]] = 1 - E[p(\mathbf{x})] = 1 - \eta_3$

$$E[\partial\psi(\mathbf{w}, \tau, \eta)/\partial\eta_3] = E\left[-\frac{1[d=1](y-\eta_1)}{\eta_3^2} - \frac{1[d=0](y-\eta_2)}{(1-\eta_3)^2}\right] = 0 - 0$$

★ since $E[1[d=1](y-\eta_1)] = E[y_1 | \mathbf{x}] - \eta_1 = 0$ (via law of iterated expect.)

★ and $E[1[d=0](y-\eta_2)] = E[y_0 | \mathbf{x}] - \eta_2 = 0$ (via law of iterated expect.)

Orthogonalization and DRL: Third input, Cross Fitting

- Also in this case, as for several causal ML methods, **we can use of independent samples for estimating the model parameters of the nuisance "parameters" (functions, also defined plug-ins) $p(X)$, $\mu_0(X)$ and $\mu_1(X)$ on the one hand and of the treatment effects, e.g. ATE on the other hand.**
- To this end, we randomly divide the total sample into (at least!) two (non-overlapping) parts or folds.
- We then estimate the model parameters for $p(X)$, $\mu_0(X)$ and $\mu_1(X)$, e.g. the coefficients in LASSO regressions (of $p(X)$, $\mu_0(X)$ and $\mu_1(X)$ on $X^{\text{First Fold}}$) of the treatment and outcome, in the first fold.
- Once they are estimated in the first fold we do not use directly these parameters! Instead, based on these parameters (e.g. coefficients), we predict in the second fold the plug-ins $p(X)$, $\mu_0(X)$ and $\mu_1(X)$ (i.e. we regress $\hat{p}(X)$, $\hat{\mu}_0(X)$ and $\hat{\mu}_1(X)$ on $X^{\text{Second Fold}}$) and estimate the treatment effect of interest (by sample analog of the population moment condition as explained before).

Orthogonalization and DRL: Third input, Cross Fitting

- **Sample-splitting prevents a correlation of the two estimation steps**, namely estimating the model parameters of the plug-ins and the treatment effect, which avoids so-called overfitting bias.
- However, this comes with the cost that only part of the data are used to estimate the causal effect, thus increasing the variance.
- We can tackle this issue by so-called cross-fitting, which consists of swapping the roles of the folds used for estimating the plug-in models and the treatment effect.

Orthogonalization and DRL: Third input, Cross Fitting

- That is, in a second estimation round, we use the second fold for obtaining the plug-in models and the first fold for treatment effect estimation. We can then ultimately estimate the ATE by taking the average of the ATE estimates in either fold.
- **How does sample-splitting (and cross-fitting specifically) prevent overfitting?** Say we used all data to estimate $\hat{\eta}$ and re-used the same data to estimate the $\hat{\alpha}$ such that $E[\psi(w_i, \hat{\alpha}, \hat{\eta})] = 0$. Then if we add a data point outside of the dataset, w_i^{test} it is likely that $E[\psi(w_i^{test}, \hat{\alpha}, \hat{\eta})] \neq 0$ (unstable);
- **How does sample-splitting (and cross-fitting specifically) prevent p-hacking?** In simple words is a matter of risk diversification: by taking a unique dataset and conducting hypotheses testing multiple times on it we risk that if any randomness occurs between X and D such that the treatment seems by chance more (less) effective, this risk is overlooked by using a unique data source. Instead by rotating and using multiple samples this risk mitigates!

Orthogonalization and DRL: Third input, Cross Fitting

- Another advantage of double robust (machine) learning is that it allows the LASSOs to select more variables in the models. In other words, the sparsity assumption, crucial for the validity of the estimators, allows only a few variables to be selected.
- **Double machine learning techniques relax this sparsity requirement** to some extent such that more variables can be selected and the estimation results are still valid.

Orthogonalization and DRL: Third input, Cross Fitting

- Importantly, under sufficiently ‘well-behaved’ plug-in estimators, **treatment effect estimation based on cross-fitting is $\sqrt{N}$ -consistent and asymptotically normal.**
- More precisely, $\sqrt{N}$ -consistency is satisfied if (among some other conditions) the plug-in estimates of $p(X)$, $\mu_0(X)$ and $\mu_1(X)$ converge to their respective true values at a rate of at least $n^{(-1/4)}$.
- This rate is slower than the conventional $\sqrt{N}$ that can be attained by many machine learning algorithms (Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey, and Robins, 2018).

Oracle property

- A property, which is worth underlying, is that under the above conditions, the **asymptotic variance of the estimators that satisfy the orthogonalization condition is unaffected by the machine learning and cross-fitting steps.**
- In other words, it is not higher than if the covariates X to be controlled for had been known a priori, rather than learned in the data by machine learning.
- This is the so-called oracle property, under which **we can compute standard errors by conventional asymptotic approximations without adjustment for the cross-fitting and machine learning steps in large enough samples.**
- So: 1) we are insured of $\sqrt{N}$ consistency if plug-ins estimates are decently estimated; 2) we are insured that we can conduct standard inference without adjusting for ML and cross-fitting via oracle property.

(Partialling-out Vs DR) + ML

- Until now we have focused on estimators of the ATE.
- As we have seen above (e.g. PLR), some methodologies **start from the assumption that we have a constant or homogeneous CATE across X which therefore coincides with the ATE.**
- **Combining DR with ML** is instead more promising from the point of view of the identification of heterogeneous treatment effects.
- Indeed **we have used the unconditional average of the machine learning-based estimates of the so-called efficient influence function $\phi(X)$ to estimate the ATE of a binary treatment, we could think about using its conditional average given X to estimate the CATE: $E[\phi(X)|X = x]$.**

Effect Heterogeneity: Intro

- In many empirical contexts, indeed, researchers and analysts are interested in whether causal effects differ importantly, i.e. are **heterogeneous**, across specific **subgroups** that can be described in terms of observed characteristics, i.e. covariates X .
- For instance, we might want to know whether the effectiveness of a **training program** or a **marketing campaign** differs across characteristics like gender, age, income, or other variables.
- Learning for which groups a specific treatment is particularly effective can help improve **treatment allocation**, e.g. for **targeting those customer** groups among which a marketing campaign entails particularly high effects on sales.

Effect Heterogeneity: Intro

It is important to notice that

- **DRL** can estimate the CATE efficiently if nuisance parameters and propensity scores are accurately estimated (just plug them in $\phi(X)$ and **CATE** $:= E[\hat{\phi}(X)|X]$ (see later slides)
- However, DRL typically relies on **predefined parametric forms** (i.e. $\phi(X)$ depending on $\mu(X)$ and η), which may not capture complex interactions or non-linear relationships between covariates and treatment effects.
- **Non-parametric methods** like **Causal Trees** (see later on) and **GRF** offer greater flexibility:
 - These methods **do not assume a specific functional form for CATE** (whereas DRL assumes the parametric form $\phi(X)$ above).

This is why such methods are needed and Causal machine learning (CML) developed!

Effect Heterogeneity: Intro

- CML combined with sample splitting can be fruitfully applied to 1) avoid making parametric assumptions on CATE; 2) investigate treatment effect heterogeneity across X , while at the same time **avoiding inferential issues of multiple hypothesis testing** when snooping for subgroups with significantly different effects.
- The latter type of issues arise when researchers look at how a treatment affects different subgroups (like age, gender, income, etc.). In this case, they often check many subgroups and then report only the treatment effects that show significant differences.
- However, **if they test too many subgroups, they may find "significant" differences just by chance**, even if those differences aren't real in the overall population. This happens because **standard statistical methods (like t-tests and p-values) are designed to test one hypothesis at a time, and they need to be adjusted when testing many hypotheses at once to avoid these random "false positives."**

Effect Heterogeneity: Intro

- Another way to see this follows. The traditional approach –that splits the sample into groups (e.g. via variable interactions) to assess the significance of the difference in the treatment effects of the groups (i.e. difference with the baseline group)– is prone to **overfitting** (e.g. if we split into smaller and smaller subgroups by introducing too many interactions);
- With rich baseline surveys, there are many, **many potential ways to split the sample**. Therefore, when looking at **many different splits** and finding a few statistically significant differences in the treatment effects, there is a good chance that the difference is **just random noise**.
- Also, other researchers might think you just tried different sample splits until you find interesting (i.e., publishable) results — of course, you would never **go fishing**... but the point is that you could do it.

Effect Heterogeneity: Intro

- **Registering a pre-analysis plan** partly solves this problem but is not entirely satisfying. You often learn new information on how the treatment could vary during data collection and you would like to be able to use these insights in analysis rather than be limited to pre-registered subgroups. In other terms, you are never sure to catch all the relevant splits ex-ante!
- Even if you pre-register your analysis successfully, traditional sample splits have another disadvantage: they typically don't **capture non-linear, higher-order interactions between treatment and baseline characteristics**.
- Put differently, the most and least affected individuals might not be individuated by one characteristic (e.g., young vs. old) but the most affected group might be a segment of the population defined by many different characteristics (in a non-linear way).

Effect Heterogeneity: Intro

This motivates the application of appropriately designed machine learning approaches for analyzing effect heterogeneity:

- to avoid **overfitting/p-hacking** (via cross-fitting/ sample splitting),
- to avoid **tying ones' hands** before going to the data,
- to identify interactions between the X s/ that define **more affected groups** of individuals.
- or, more humbly, to identify a variable as a relevant dimension of heterogeneity in a way that is **robust to possible interactions effects with other variables**

Notice: CATE is the estimator of effect heterogeneity **if the latter is present**. If no effect heterogeneity is present then $CATE=ATE$ as we will see.

Causal Tree

- Let us first consider the case that our **treatment D is randomly assigned** as in successful experiments such that covariates X are not required to make subjects comparable across treatment states when assessing causal effects, but may nevertheless be exploited to assess effect heterogeneity (**i.e. X are not needed to satisfy CIA but only to check for the presence of effect heterogeneity**).
- In this context, Athey and Imbens (2016) suggest **our first CML algorithm**: a causal tree method that builds on two key modifications of conventional decision trees for prediction.
- With respect to traditional machine-learning trees built that predict Y , in causal trees the aim is to predict **the mean difference in Y across treatment groups**. **In other words, $Y(1) - Y(0)$ serves as the outcome to be predicted when recursively splitting the covariate space into subgroups**. So splits are based on $Y(1) - Y(0)$.
- Of course **the problem is that $Y(1) - Y(0)$ is unobserved** (so e.g. no out-of-sample GoF available).

Causal Tree

- Causal trees aim to group individuals into subgroups based on covariate values to understand how the treatment effect varies across these groups.
- The algorithm thus aims at generating covariate value-specific subgroups in a way that minimizes the sum of squared residuals in effect ($Y(1) - Y(0)$) estimation (rather than outcome prediction).
- The goal is to find subgroups where the treatment effect is as similar as possible within each subgroup (maximize effect homogeneity) but can vary across different subgroups.
- The way a causal tree is built creates groups where the treatment effects are as different as possible between groups (most heterogeneous treatment effects), based on the number of splits (divisions) the tree makes. So, each split in the tree tries to separate individuals into groups that respond differently to the treatment, allowing us to see distinct differences in treatment effects across these groups

Causal Tree

- As a second modification with respect to the off-the-shelf tree used for prediction, a causal tree applies **sample splitting to use different parts (or folds) of the data for estimating (a) the tree's structure (a sort of nuisance parameters if we want a comparison with DRL) and (b) the treatment effects within subsets.**
- This avoids spuriously large effect heterogeneities due to overfitting, but the results tend to be very sensitive to the set of variables included in the model and to the hyperparameters chosen. In other words, sample-splitting prevents that –when trying to estimate the CATE for a new observation using the trained tree– the CATE of the latter observation is completely missed and the observation is classified in a wrong group; **however it does not prevent that, adding a covariate more, say X_k , the structure of the new tree with X_k changes a lot wrt the previous tree (high variance).**
- We can apply the causal tree method also when the **treatment is not randomly assigned**, but the selection-on-observables assumptions hold for a pre-selected (rather than machine learning-selected) set of covariates.
- In this case, we may **reweight outcomes by the inverse of the propensity score** before taking differences in treated and nontreated outcomes and applying the causal tree approach.

Interested in deepening causal trees formally? If we have time and you are interested we can take a look to the Appendix!

A cool extension of causal trees are causal random forests and GRF. In the Appendix we talk a little about them!

Testing for the presence of effect heterogeneity

- Causal trees (forests) directly estimate CATE non-parametrically;
- In a sense, causal trees **assume the presence of effect heterogeneity**;
- However, it is not always the case that effect heterogeneity is present and we would like to test formally whether causal effects differ importantly (i.e. are heterogeneous) across specific subgroups described by X ;
- Let's not be confused: we are testing the heterogeneity of CATE (i.e. if CATE effectively represents heterogeneous effects). Indeed CATE is a generalization of ATE. If no heterogeneous effect is identified, then simply "homogeneous CATE" = ATE!
- To do so, **let's come back to our parametric approach since this testing has been formalized before by Semenova and Cherozokov (2021) in such a scenario** with the $\phi(X)$ influence function

Testing for the presence of effect heterogeneity

- Specifically learning about the heterogeneity of the CATE is based on **the combined use of the Doubly Robust Estimator with ML**.
- The machine learning-based estimate of the **efficient influence function** $\phi(X)$ offers, as we have seen, the possibility to estimate the ATE of a binary treatment just as the (unconditional) average of this function $E[\phi(X)]$.
- More importantly here, its conditional average given X yields the CATE: $E[\phi(X)|X]$.
- This suggests that we may investigate effect heterogeneity by **regressing the estimated efficient influence function $\hat{\phi}(X)$ on X or a subset of the covariates**⁸, as for instance an indicator for gender if effect heterogeneity across gender is of interest.
- Notice indeed that $\phi(X)$ is **unknown** and **MUST** be estimated before either via DRL as above or directly via some ML method.

⁸Why? Because CATE takes the form of a conditional function (remember in OLS we regress $E[Y|X]$ on covariates. Here the Y is considered $\phi(X)$)

Testing for the presence of effect heterogeneity

- When $\phi(X)$ estimated via DRL...
- Then, as discussed in Semenova and Chernozhukov (2021)⁹, running a OLS regression of $\hat{\phi}(X)$ on a limited number of pre-selected covariates¹⁰ can, under certain conditions, yield **asymptotically correct coefficient estimates and standard errors** for testing whether effect heterogeneity is statistically significant (i.e. if CATE varies effectively with some X s).
- Two important **conditions** are that 1) the ML estimate $\hat{\phi}(X)$ of $\phi(X)$ is done via **cross-fitting**, i.e. in a different data fold than the conditional mean outcome and the propensity score (i.e. we learn propensity score and conditional mean outcome on the first fold and we then predict the $\hat{\phi}(X)$ – learned by plugging-in the learned conditional mean outcome and the propensity score in First Fold – on different second fold data, see above), and 2) that the **plug-in parameters –learned by predicting $\hat{\phi}(X)$ in the Second Fold– are estimated at least at a $n^{(-1/4)}$ rate** (i.e. they can suck but not that much). These are satisfied via DRL as seen.

⁹Ask me if you want more formal identification results. See Appendix.

¹⁰Semenova (2022) regularizes with LASSO

Testing for the presence of effect heterogeneity

- This implies that, for instance, one can infer from the coefficient estimate on a gender indicator and its standard error whether the CATE statistically significantly differs across gender (and similarly so for other covariates).
- This is remarkable, because unlike in a standard OLS regression with a known or directly observed outcome, $\phi(X)$ needs to be estimated by machine learning first.
- But similarly as for ATE estimation averaging unconditionally $\phi(X)$, **the machine learning step does under certain conditions not affect the asymptotic behavior of the OLS regression of the ML-estimated $\hat{\phi}(X)$ on X such that we can assess effect heterogeneity with $\sqrt{N}$ -consistency** (i.e. oracle property is valid also on the coefficients of CATE heterogeneity).

Testing for the presence of effect heterogeneity (continuous case)

- When investigating effect heterogeneity across continuously distributed (rather than binary or discrete) covariates as for instance income, we might want to avoid imposing a linear association between effect heterogeneity and the covariates.
- Why? For binary or discrete variables (like "male/female" or "low/high income"), a linear or simple model is less problematic because there are only a few categories to compare. In these cases, we're just looking at average differences between groups. But with continuous variables like income, the range of values is broader, and there's more room for nuanced variations that a straight line would ignore.
- For this reason, we may prefer a **nonparametric kernel regression of $\phi(X)$ on the continuous covariates** instead of linear regression.
- Also in this case, the machine learning-step does not affect the convergence rate and asymptotic behavior of the kernel regression if certain conditions are met (Zimmert and Lechner (2019); Fan, Hsu, Lieli, and Zhang, 2020). For instance, if we only considers a single continuous covariate, one condition is that the product of the convergence rates of the plug-in parameters must not be slower than $n^{(-3/5)}$.

Testing for the presence of effect heterogeneity (continuous case)

- Rather than considering a limited number of pre-selected covariates for CATE estimation, **we may also be interested in detecting the covariates that most importantly predict effect heterogeneity in a data-driven way.**
- Notice: (i) W.r.t. before where we run an OLS on $\hat{\phi}(X)$, i.e. investigating the causal relations of X on CATE (i.e. which variables cause the CATE), here we perform a prediction task (which X predict the CATE the best); (ii) This is particularly useful where we have tons of variables such that $\text{Nr. variables} \geq N$ so that OLS of $\hat{\phi}(\cdot)$ cannot be performed and therefore we need to have an idea of which X to include!
- To this end, **we might in principle apply a machine learning algorithm to predict directly $\phi(\cdot)$ as a function of X and assess which covariates have the highest predictive power.** Hence we are in the case **when $\phi(\cdot)$ is estimated directly via ML...**
- In a LASSO regression of $\phi(X)$ on the covariates (and possibly interaction and higher order terms), the standardized coefficients (standardized based on the standard deviation of the covariates) permit ranking the covariates according to their importance in terms of effect heterogeneity.

Testing for the presence of effect heterogeneity (inference)

- However, conducting inference (e.g. computing p-values and confidence intervals on the goodness of the estimated coefficients via LASSO) or hypothesis tests about the importance of covariates for effect heterogeneity is less straightforward and generally requires further sample splitting steps.
- This is due to the threat of overfitting: i.e. we select important covariates by machine learning and assess their statistical significance in the very same data.
- In other words: "When we use LASSO on a dataset, it picks the variables that best fit that specific data. If we then test how important these chosen variables are using the same data, it's like "double-dipping"—we're reusing data that's already influenced by LASSO's choices. This can make us too confident in the importance of some variables, leading to biased results and underestimating the true uncertainty."
- This implies that we should use different folds of the data for discovering and selecting the most influential covariates on the one hand and for statistical inference (e.g. assessing whether the selected covariates statistically significantly determine effect heterogeneity) on the other hand (see for example the algorithm suggested in **Lee, Bargagli-Stoffi, and Dominici, 2020**; as well as the sample splitting approach in Athey and Imbens, 2016).

STATA and Code

LASSO to select covariates and DR

- In Stata this is **implemented by telasso**
- The telasso estimator uses a five-step approach to estimating ATE:
 - 1 It uses lasso techniques to select variables in the outcome model for each treatment level (solves CIA).
 - 2 Based on the selected variables in step 1, it fits separate regression models of the outcome for each treatment level and obtains the treatment-specific predicted outcomes for each subject.
 - 3 It uses lasso techniques to select variables in the treatment model (solves overlap)
 - 4 Based on the selected variables in step 3, it estimates the parameters of the treatment model and computes inverse-probability weights.
 - 5 It computes the weighted means of the treatment-specific predicted outcomes, where the weights are the inverse-probability weights computed in step 4. The contrasts of these weighted averages provide the estimates of the ATE.

- **Step 1: Lasso to Select Variables $\tilde{X}$ for Outcome Models**

$$Y_i = X_i\beta_1 + \varepsilon_i \quad \text{for } D_i = 1, \quad Y_i = X_i\beta_0 + \varepsilon_i \quad \text{for } D_i = 0.$$

- **Step 2: Fit Separate Outcome Models**

$$\hat{Y}_1(\tilde{X}_i) = \tilde{X}_i\hat{\beta}_1, \quad \hat{Y}_0(\tilde{X}_i) = \tilde{X}_i\hat{\beta}_0.$$

- **Step 3: Lasso to Select Variables for the Treatment Model**

$$p(X_i) = \Pr(D_i = 1 \mid X_i) = \frac{e^{X_i\gamma}}{1 + e^{X_i\gamma}}.$$

- **Step 4: Estimate the Treatment Model and Compute Weights**

$$w_i = \frac{1}{p(\tilde{X}_i)} \quad \text{if } D_i = 1, \quad w_i = \frac{1}{1 - p(\tilde{X}_i)} \quad \text{if } D_i = 0.$$

- **Step 5: Compute Weighted Averages and Estimate ATE**

$$\hat{E}[Y_1 \mid D = 1] = \frac{1}{N_1} \sum_{i:D_i=1} w_i \cdot \hat{Y}_1(\tilde{X}_i), \quad \hat{E}[Y_0 \mid D = 0] = \frac{1}{N_0} \sum_{i:D_i=0} w_i \cdot \hat{Y}_0(\tilde{X}_i).$$

$$\text{ATE} = \hat{E}[Y_1 \mid D = 1] - \hat{E}[Y_0 \mid D = 0].$$

LASSO to select covariates and DR

- Steps 1 and 3 perform the model selection for the outcome and treatment models, respectively.
- Using the selected variables, steps 2, 4, and 5 construct moment conditions to estimating ATEs.
- The resulting estimator is consistent under CIA and overlap.
- The inference is robust to the mild model selection mistakes that could happen in steps 1 and 3.
- This estimator is also robust to model misspecification in either the outcome or the treatment model because of the double robust moment condition used in step 5.
- see `LASSO_for_inference.do` for an example

Example in R (ML_causality.R)

- To illustrate the implementation of the DR+ML strategy in R, we load the causalweight package in R and use the JC data from for the evaluation of the Job Corps program.
- It is an experiment conducted between November 1994 and February 1996 which randomized access to Job Corps, a large U.S. education program financed by the U.S. Department of Labor that targets disadvantaged individuals aged 16 to 24.
- In our empirical example, we aim at assessing the average treatment effect of we are interested in the effect of training participation in the first year after program assignment on the health status four years after assignment when controlling for the baseline covariates measured prior to training.
- The treatDML command that we will use from this package by default applies lasso regression for the estimation of the propensity scores and conditional mean outcomes.
- We will also check what happens by using the random forest algorithm.

Appendix

Orthogonalization and DRL: when applicable?

In the following we will provide a general version of the DR problem described in the slides.

Specifically, DRL is applicable in models taking the following general form:

$$Y_{(d)} = g_d(X, W) + \epsilon_d \quad \mathbb{E}[\epsilon_d | X, W] = 0, \quad (1)$$

$$\Pr[D = d | X, W] = p_d(X, W), \quad (2)$$

$$\{Y_d\}_{d=1}^{n_d} \perp D \mid X, W. \quad (3)$$

- X and W high-dimensional
- No further structural assumptions are made on g_d and p_d .
- Both are estimated non-parametrically using machine learning methods.
- The CATE for $d = 0$ is defined as:

$$\theta_d(X) = \mathbb{E}[Y_{(d)} - Y_{(0)} \mid X] = \mathbb{E}[g_d(X, W) - g_0(X, W) \mid X].$$

- As seen, the DR approach combines two methods: direct regression and IPW to debias the model.
- Another angle to approach DR is to see it as estimating the potential outcomes as follows:

$$Y_{i,d}^{DR} = g_d(X_i, W_i) + \frac{Y_i - g_d(X_i, W_i)}{p_d(X_i, W_i)} \cdot 1\{D_i = d\}$$

- The CATE $\theta_d(X)$ can be learned by regressing $Y_{i,d}^{DR} - Y_{i,0}^{DR}$ on X_i .
- The algorithm steps are the following described in the recipe on the main text but for general functions:
 - Learn a regression model $\hat{g}_d(X, W)$ by regressing Y on D, X, W .
 - Learn a propensity model $\hat{p}_d(X, W)$ by classifying D based on X, W .
 - Construct the doubly robust variables and regress them on X .
- Advantage: The final estimate of $\theta_d(X)$ depends on the product of the mean squared errors of the regression and propensity models. As long as one of these models is accurate, the estimate is robust.

Semenova (2021) vs $\phi(X)$ on X

You may have seen Semenova (2021) as regressing residuals of Y on residuals of D . However, regressing $\phi(X)$ on the relevant X yields equivalent results: **Goal:** Estimate treatment heterogeneity based on key covariates, e.g., age and income (i.e. assuming that age and income are the only relevant covariates individuated correctly by the two approaches.).

Moreover $E[\phi(X)|X] = D'_{it}\beta_0$ in Semenova (2021).

- **Semenova's Approach:**

- Residualize Y and D to get $\tilde{Y}$ and V .
- Estimate β_0 in $\tilde{Y} = V'\beta_0 + \text{error}$ using regularization.

- **Simplified Approach:**

- Compute $\phi(X)$ (an estimator for treatment effect).
- Run OLS (call it OLS 1): $\phi(X) = \gamma_1 \cdot \text{age} + \gamma_2 \cdot \text{income} + \text{error}$.

- **Equivalence:** If age (X_1) and income (X_2) capture the main sources of heterogeneity, both methods yield similar insights into treatment effect variation across these covariates:

- $\tilde{Y} = Y - l(X_1, X_2)$, assuming $l(\cdot)$ linear then: $\tilde{Y} = Y - \delta_1 X_1 - \delta_2 X_2$;
 $V = D - g(X_1, X_2)$, if $g(\cdot)$ linear, $V = D - \alpha_1 X_1 - \alpha_2 X_2$
- Regress $\tilde{Y}$ on V to get $\hat{\beta}_0 = (V'V)^{-1} V'\tilde{Y}$
- Rewriting OLS 1 in compact form: $\phi(X) = X'\gamma + \text{error}$ and OLS of γ is
 $\hat{\gamma} = (X'X)^{-1} X'\phi = (X'X)^{-1} X'D'_{it}\beta_0$. Then γ is essentially an estimate of β_0 projected onto the space spanned by X . Hence γ is essentially an estimate of β_0 projected onto the space spanned by X and D which is equivalent to say that γ is the result of regressing β_0 on the relevant X s and D .
- In other words, since $D'_{it}\beta_0$ is the true CATE, i.e. $E[\phi(X)|X]$, since OLS is efficient and consistent when regressing functions of the type $E[Y|X]$ on X , since testing heterogeneity on X means to identify the X s that generate heterogeneity and since γ is the result of regressing β_0 on the relevant X s and D , we can conclude that regressing $D'_{it}\beta_0 := E[\phi(X)|X]$ on X identifies heterogeneous patterns.

Causal trees

The main difficulty, as we know, is that, differently from off-the-shelves trees where, being a supervised technique, the target Y is observed, in causal trees we do not observe the target $Y(1) - Y(0)$.

How does this impact? It basically impacts on the construction of the in and out of sample Goodness of Fits (GoF) functions Q^{in} and Q^{out} . This is important because as we know the functional that allows us to decide to split or not a leaf into children is based on Q^{crit} which is a simple transformation of Q^{is} ! The GoF indeed assume that the target is observed!

Let's introduce a general algorithm for building trees that for the moment ignores that we do not observe the target $Y(1) - Y(0)$ (i.e. suppose that we have a splitting criterion somehow Q^{crit}).

We also divide the algorithm in two parts for the sake of simplicity:

First part of Algorithm

1. Fix α , the tuning parameter for penalty term. Initialize tree $\mathcal{T}$ with the root node containing $\{(1, X)\}$. Set of completed nodes $\mathcal{C} = \emptyset$.

2. **Iterative Process: Until $\mathcal{T}^{\text{term}}$ is fully processed**

- For each terminal node $t \in \mathcal{T}^{\text{term}}$ (not in $\mathcal{C}$):

- (a) Construct estimator $\hat{\tau}(\cdot; \mathcal{T})$ as follows: for each terminal node $t \in \mathcal{T}^{\text{term}}$ denoted X_t , estimate $\hat{\tau}(\cdot; \mathcal{T})$ as a constant for all $x \in X_t$.

- (b) **For each feature $l = 1, \dots, L$:**

- For each potential threshold x_l^{thr} : Create candidate tree $\mathcal{T}_{x_l^{\text{thr}}}$ by splitting X_t into two nodes $2t$ and $2t + 1$ based on x_l^{thr} :

$$\{x \in X_t : x_l \leq x_l^{\text{thr}}\} \text{ and } \{x \in X_t : x_l > x_l^{\text{thr}}\}$$

- Generate new estimate $\hat{\tau}(\cdot; \mathcal{T}_{x_l^{\text{thr}}})$.

- Find optimal split $x_l^{*, \text{thr}}$ that maximizes $Q^{\text{crit}}(\hat{\tau}(\cdot; \mathcal{T}_{x_l^{\text{thr}}}); \alpha, X^{\text{tr}}, Y^{\text{tr, obs}})$.

- (c) **Update Terminal Nodes:**

- If $\max_{l=1}^L Q^{\text{crit}}(\cdot; \mathcal{T}_{x_l^{*, \text{thr}}}) \leq Q^{\text{crit}}(\hat{\tau}(\cdot; \mathcal{T}); \alpha, X^{\text{tr}}, Y^{\text{tr, obs}})$, add t to $\mathcal{C}$.

- Otherwise, select feature l^* with highest gain and update $\mathcal{T}$ to $\mathcal{T}_{x_{l^*}^{*, \text{thr}}}$.

3. **Finalize Tree Structure**

- Define T^α as the final tree after iteration, and $\hat{\tau}^\alpha(x)$ as the associated estimator.

Try to simplify the first part of the algorithm. Say we fixed α to some value and we have 3 features: X_1, X_2, X_3 .

For each of them define k possible thresholds:

- $x_1^{thr,1}, x_1^{thr,2}, x_1^{thr,3}, \dots, x_1^{thr,k}$ for X_1
- $x_2^{thr,1}, x_2^{thr,2}, x_2^{thr,3}, \dots, x_2^{thr,k}$ for X_2
- $x_3^{thr,1}, x_3^{thr,2}, x_3^{thr,3}, \dots, x_3^{thr,k}$ for X_3

For each threshold of each X compute $\hat{\tau}$ and Q^{crit} . Example for X_1 follows:

$$\begin{array}{c} \nearrow \hat{\tau}_{X_1 \leq x_1^{thr,1}}(\cdot) \\ t \\ \searrow \hat{\tau}_{X_1 > x_1^{thr,1}}(\cdot) \end{array}$$

$$\begin{array}{c} \nearrow \hat{\tau}_{X_1 \leq x_1^{thr,2}}(\cdot) \\ t \\ \searrow \hat{\tau}_{X_1 > x_1^{thr,2}}(\cdot) \end{array}$$

, ...,

$$\begin{array}{c} \nearrow \hat{\tau}_{X_1 \leq x_1^{thr,k}}(\cdot) \\ t \\ \searrow \hat{\tau}_{X_1 > x_1^{thr,k}}(\cdot) \end{array}$$

Compute the $x_1^{t,*}$ maximizing $Q^{crit}(\cdot)$.

Repeat the same for the thresholds of X_2 and X_3

So we end up with three (nr. features) thresholds maximizing $Q^{crit}(\cdot)$:
 $x_1^{t,*}, x_2^{t,*}, x_3^{t,*}$

We need now to compare the maximum among
 $Q^{crit}(\cdot; \hat{\tau}(\mathcal{T}_{x_1^{t,*}})), Q^{crit}(\cdot; \hat{\tau}(\mathcal{T}_{x_2^{t,*}})), Q^{crit}(\hat{\tau}(\cdot; \mathcal{T}_{x_3^{t,*}}))$; say the threshold that
maximizes the criterion is $x_2^{t,*}$ and call $Q^* := Q^{crit}(\cdot; \hat{\tau}(\mathcal{T}_{x_2^{t,*}}))$ and
compare it with the $Q()$ that we had, if we did not made any split,
 $Q^{crit}(\hat{\tau}(\cdot, \mathcal{T}))$ call it Q^{orig}

Two possible scenarios: 1) IF $Q^* > Q^{orig}$ add leaf to t and create the new
tree with the threshold at $x_2^{t,*}$ forming two more terminal nodes (not in $\mathcal{C}$,
i.e. we need to re-evaluate them with the above procedure); IF
 $Q^* \leq Q^{orig}$, then add t to $\mathcal{C}$.

Notice that in reality, as we will see below the algorithm employs a transformation of Y_i , namely $Y_i^* = \frac{D_i - p}{p(1-p)} Y_i$, i.e. is a sort of IPW correcting for the fact that, within each leaf the proportion of treated may not represent the true proportion of treated within the population. Importantly, when $D_i = 1$, $Y_i^* = \frac{1}{p} Y_i$ and if $D_i = 0$, $Y_i^* = -\frac{1}{1-p} Y_i$ such that $E[Y_i^*] = E[\tau_i]$. Hence, $E[Y_i^* | X = x] = \tau_i$

Usually, $\tau(\cdot)$ is the sample mean of Y_i^* within leaf and the set of candidate estimators correspond to different specifications of how the tree is split (e.g. see above $Q^{crit}(\cdot)$: $x_1^{t,*}, x_2^{t,*}, x_3^{t,*}$).

For a fixed tuning parameter λ , $Q^{is} := -MSE := -\frac{1}{N} \sum_{i=1}^N (\hat{\tau}_i^* - Y_i^*)^2$ such that $Q^{crit} := Q^{is} - \lambda \times \# \text{ leaves}$

Usually we perform cross-validation on a grid of tuning parameters and choose the λ^* with highest $Q^{os} = -MSE$ (with test data).

Second part of Algorithm (choose optimal α with R-fold validation)

Take a set A of possible α s the lower of which is α_0 ;

1. Partition Training Sample

- Partition the training sample into R subsamples: $\{X_{(r)}^{\text{tr}}, Y_{(r)}^{\text{tr,obs}}\}$ for $r = 1, \dots, R$.
- Define $\hat{\tau}_{(r)}^{\text{pru}}(\cdot; \alpha)$ iteratively for each subsample.

2. Build and Prune Large (i.e. with lowest penalty α_0 on complexity) Tree $\mathcal{T}^{\alpha_0, r}$.

- Build a large tree $\mathcal{T}^{\alpha_0, r}$ using $(X_{(r)}^{\text{tr}}, Y_{(r)}^{\text{tr,obs}})$.
- Initialize $\mathcal{T}^{\text{pru}, r} = \mathcal{T}^{\alpha_0, r}$ and set $u = 1$.

3. Iterative Pruning Process

- Repeat until $\mathcal{T}^{\text{pru}, r} = \{(1, X)\}$:
 - For each node t in $\mathcal{T}^{\text{pru}, r}$, define subtree $\mathcal{T}_{(-t)}^{\text{pruned}, r}$ by deleting all children of node t .
 - Calculate

$$\Delta(t, \mathcal{T}^{\text{pru}, r}) = \frac{Q^{\text{is}}(\hat{\tau}(\cdot; \mathcal{T}^{\text{pru}, r}); X_{(r)}^{\text{tr}}, Y_{(r)}^{\text{tr,obs}}) - Q^{\text{is}}(\hat{\tau}(\cdot; \mathcal{T}_{(-t)}^{\text{pru}, r}); X_{(r)}^{\text{tr}}, Y_{(r)}^{\text{tr,obs}})}{|\mathcal{T}^{\text{pru}, r}| - |\mathcal{T}_{(-t)}^{\text{pru}, r}|}$$

- Find the "weakest link" t^* that maximizes $\Delta(t, \mathcal{T}^{\text{pru}, r})$.
 - For $\alpha \in [\alpha_{u-1}, \Delta(t^*, \mathcal{T}^{\text{pru}, r})]$, define $\hat{\tau}_{(r)}^{\text{pru}}(\cdot; \alpha) = \hat{\tau}(\cdot; \mathcal{T}_{(-t^*)}^{\text{pru}, r})$.
 - Increment $u = u + 1$.
- For $\alpha \in A$ such that $\alpha > \alpha_{u-1}$, set $\hat{\tau}_{(r)}^{\text{pru}}(\cdot; \alpha) = \hat{\tau}(\cdot; \{(1, X)\})$.

4. Goodness-of-Fit Evaluation

- Evaluate the goodness-of-fit on each subsample using the choice of Q^{os} .
- Average the goodness-of-fit across subsamples:

$$\bar{Q}^{\text{os}}(\alpha) = \frac{1}{R} \sum_{r=1}^R Q^{\text{os}}(\hat{\tau}_{(r)}^{\text{pru}}(\cdot; \alpha); X_{(r)}^{\text{tr}}, Y_{(r)}^{\text{tr,obs}})$$

5. Select Optimal α

- Choose α^* that maximizes the criterion:

$$\alpha^* = \arg \max_{\alpha \in A} \bar{Q}^{\text{os}}(\alpha)$$

6. Define Final Estimator

- Let $\mathcal{T}^*$ be the optimal tree $\mathcal{T}^{\alpha^*}$ using the full sample with $\alpha = \alpha^*$.
- Final estimator is $\hat{\tau}^{\alpha^*}(x) = \hat{\tau}^{\alpha^*}(x)$.

A quick note:

To identify the "weakest link" t^* for pruning in a decision tree $\mathcal{T}^{\text{pru},r}$, we maximize the criterion

$$\Delta(t, \mathcal{T}^{\text{pru},r}) = \frac{Q^{\text{is}}(\hat{\tau}(\cdot; \mathcal{T}^{\text{pru},r}); X_{(r)}^{\text{tr}}, Y_{(r)}^{\text{tr,obs}}) - Q^{\text{is}}(\hat{\tau}(\cdot; \mathcal{T}_{(-t)}^{\text{pru},r}); X_{(r)}^{\text{tr}}, Y_{(r)}^{\text{tr,obs}})}{|\mathcal{T}^{\text{pru},r}| - |\mathcal{T}_{(-t)}^{\text{pru},r}|}.$$

In this formulation:

- The numerator, $Q^{\text{is}}(\hat{\tau}(\cdot; \mathcal{T}^{\text{pru},r})) - Q^{\text{is}}(\hat{\tau}(\cdot; \mathcal{T}_{(-t)}^{\text{pru},r}))$, represents the loss in in-sample prediction accuracy due to pruning node t .
- The denominator, $|\mathcal{T}^{\text{pru},r}| - |\mathcal{T}_{(-t)}^{\text{pru},r}|$, captures the reduction in model complexity achieved by pruning t .

Maximizing $\Delta(t, \mathcal{T}^{\text{pru},r})$ identifies the node t^* whose pruning yields the smallest accuracy loss relative to complexity gain, effectively finding the "weakest link" that contributes minimally to model accuracy per unit of complexity.

In other words, the second part of the algorithm chooses a "pruning" criterion for the tree α (a tree is more complex the less it is pruned);

What it's done is simply to iteratively "cut" the weakest links and evaluate how much, cutting these links is beneficial according to an out-of-bags criterion Q^{os} .

The criterion α pruning "at best" is chosen and, as a consequence, the optimal estimator $\hat{\tau}^{\alpha^*}$ is chosen.

Now the clever part of Athey et al. (2016) is basically to notice that the problem that the target $Y(1) - Y(0)$ is not observed is just a matter of adjusting only the criterion Q^{crit} to accomplish the goal and retaining the full algorithm as it is described above (which is pretty similar to the out-of-shelves trees' algorithm!)

Let's try to give an intuition of this. First of all, notice that under Rubin (1983) we can re-write the outcomes as

$$Y_i^* = Y_i^{obs} \cdot \frac{D_i - p}{p(1 - p)}$$

where p is the propensity score (equal across X s if randomization holds).

Now: this transformation simply re-weights treated by 1 and controls by $1(1 - p)$ ¹¹

The latter transformation allows to adopt the method also when the treatment is not randomly assigned (it transforms the outcomes so that they look as if the treatment were randomly assigned) as said in the main text.

Under unconfoundedness we can show then that:

$$E[Y_i^* | X_i = x] = \tau(X)$$

¹¹: just try to substitute $D_i = 1$ and $D_i = 0$

IMPORTANT NOTE: It may seem at first sight that, since everything is observed in Y_i^* we may revolve to off-the shelf machine-learning trees to estimate the CATE. After all $E[Y_i^* | X_i = x] = \tau(X)$!

However this would not be optimal! Why?

- Estimating the Conditional Average Treatment Effect (CATE) $\tau(x)$ can be done more efficiently by using the **triple** $(Y_i^{\text{obs}}, D_i, X_i)$ rather than the **pair** (Y_i^*, X_i) .
- Using only the pairs (Y_i^*, X_i) discards valuable information available in the full triple, which may lead to less efficient estimates.
- When there is no variation in treatment effects, i.e., $\text{Var}(Y_i^{\text{obs}} | D_i, X_i) = 0$, CATE can be estimated without error for all values of X_i .
- However, if there is **variation in the treatment effect** across individuals, then $\text{Var}(Y_i^* | X_i) > 0$, leading to estimation error in $E[Y_i^* | X_i]$.

WHY IS THE LATTER THE CASE???

Simply put, it can be shown that Y_i^* can be re-written as a function of $Y_i(1) - Y_i(0)$ so that when there are variations in the latter, $\text{Var}(Y_i^* | X_i) > 0$.

Differently when estimating $\text{Var}(Y_i^{\text{obs}} | D_i, X_i) = 0$, we compute the variance inside the group of treated and inside the group of controls (i.e. we condition on W_i) such that variations in $Y_i(1) - Y_i(0)$ do not affect $\text{Var}(Y_i^{\text{obs}} | D_i, X_i) = 0$.

Hence the problem remains and even if we have Y_i^* we cannot use off-the-shelves trees to compute the CATE!

From here the literature on causal trees divided into two major branches depending on the GoF for in-sample and out-of-sample adopted.

Remember that the ideal but infeasible GoF is

$$Q^{infeas} = -\frac{1}{N^{te}} \sum_{i=1}^{N^{te}} (Y_i(1) - Y_i(0) - \hat{\tau}(X_i))^2$$

which is infeasible because we do not observe $Y_i(1) - Y_i(0)$.

The first branch tries to estimate the counterfactual of Y_i^{obs} , while the second focus directly in estimating Y_i^* and the CATE as $E[Y_i^*|X_i]$, i.e. ignoring in toto the treatment indicator D_i

In the case of the most used causal tree algorithm we have that $\hat{\tau}(X)$ is represented by the difference between the weighted sample averages of treated and control in a leaf. $Q^{is} = -(Y_i^{*,tr} - \hat{\tau}(X_i^{tr}))^2$,
 $Q^{os} = -(Y_i^{*,test} - \hat{\tau}(X_i^{test}))^2$

Causal (Random) Forest

- A further and related method for investigating effect heterogeneity is the causal (random) forest, see Wager and Athey (2018) and Athey, Tibshirani, and Wager (2019).
- As the name suggests, it is a modified version of the random forest and can (in contrast to the causal tree) also be applied **when important control variables are not pre-selected but adjusted for by machine learning** as discussed before, given that CIA holds.
- The causal forest consists of several steps.
- **First, we predict both Y and D as a function of X using random forests and leave-one-out cross-validation.**

Causal (Random) Forest

- The latter implies that the outcome or treatment of each observation is predicted based on all observations in the data but its own, in order to prevent overfitting when conditioning on X .
- **Second**, the predictions are used for computing residuals of the outcomes and treatments, which corresponds to the **partialling out strategy** discussed before.
- **Third**, we **predict the effect of the residuals of D on the residuals of Y as a function of X by yet another random forest**, which averages over a large number of causal trees that use different folds of the respective tree-specific samples for (i) modeling effect heterogeneity by a tree-structure and (ii) estimating treatment effects in within the subsets.
- Put simply, this method combines the idea of sample splitting and partialling out to control for important confounders with the causal tree approach for finding effect heterogeneity, however, based on averaging over many trees.

Causal Tree Vs Causal (Random) Forest

- When comparing a single **causal tree** to a causal forest, an advantage of the former is that it directly yields **easy-to-interpret splitting rules** for defining subgroups based on the most predictive covariates in terms of effect heterogeneity.
- Therefore, also **interactions between covariates can be explicitly identified (which define the final splits)**.
- On the negative side, tree structures tend to have a **high variance** such that a small change in the data may entail quite different splitting rules.
- The **causal forest** is **more attractive in terms of variance**, but does on the other hand **not provide** straightforward guidance on how to define **subgroups** due to averaging over many trees.

Causal Tree Vs Causal (Random) Forest

- However, it yields an estimate of the $CATE_x = E[Y(1) - Y(0)|X = x]$.
- Therefore, we can **investigate the heterogeneity of the CATE as a function of the covariates X** .
- We may, for instance, split the sample into several categories with higher and lower CATE estimates, e.g. based on specific quantiles (like the median) of the CATE distribution, and investigate whether the averages of specific or all covariates differ importantly across categories.

Causal Tree Vs Causal (Random) Forest

- It is also worth mentioning that appropriately averaging over the CATE estimates in the total sample or among the treatment provides consistent estimates of the ATE and ATT, respectively!!
- Finally, the causal forest can also apply it to a continuously distributed treatment D yielding the conditional average partial effect (CAPE) of increasing the currently given treatment intensity by a small amount when keeping the covariates fixed.
- This formally corresponds to the average derivative of Y w.r.t. D given X .